

CLASSIFICATION AND MAJOR GROUP OF FISHES

PUACP

The Major Groups of Fishes

- ❑ The subject of ichthyology has been defined as the study of fishes, and fishes have been defined primarily as aquatic and gill-breathing vertebrates equipped with fins.
- ❑ Beyond this generality, the diversity of shapes, sizes, and varieties of fishes, and the different distributions of the many kinds in time and space are so great that many difficulties have been encountered in trying to classify them.
- ❑ There are 5 major groups of fish, however, each class of fish is separated into smaller taxa such as orders, families, genera and species. The major groups also contain various subgroups and suborders.

SCOPE OF CLASSIFICATION

- ❑ The materials to be classified are the specimens of all of the thousands of living and fossil kinds of fishes that have been discovered and all of those yet to become known to science.
- ❑ Almost any area of the earth may be expected to yield new examples to be described and categorized.
- ❑ As commonly conceived these fishes include some without jaws, of which a few are living and unarmored (the lampreys and hagfishes), along with others that are archaic and armored, and the many fishes with jaws, some extinct and some living (sharks and bony fishes and their relatives).

AIMS OF CLASSIFICATION

At least two Aims exist: (1) practical; and (2) ideal.

- ❑ The practical aim is simply to arrange the kinds of fishes into some system for the pure convenience of finding again something that has been handled before, such as in the alphabetical arrangement by name of specimens in jars on a shelf.
- ❑ Such a functional arrangement might also be adopted to make it as easy as possible to distinguish one kind from another.
- ❑ Thus, the basic theme of a practical classification would be for mechanical convenience-it might be done alphabetically, by size, or by any other useful means.

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- ❑ If we add to the utilitarian aspects of the practical aim, the thought that the Arrangement should also depict the evolutionary history of fishes.
- ❑ The ideal aim, then, is for classification to be evolutionary, genetic, or natural in character rather than merely artificial or mechanical.
- ❑ Gaps in knowledge, however, make it difficult to achieve a purely ideal aim.
- ❑ The result is that modern classifications of fishes, although based on ideal aims, are always partly artificial.

METHOD OF CLASSIFICATION

Simultaneous attainment of practical and ideal objectives in the classification of plants and animals, including fishes, early led to the acceptance of the following seven standard categories:

1. Kingdom
2. Phylum
3. Class
4. Order
5. Family
6. Genus
7. Species

These are listed from largest (most inclusive) to smallest, Hence also from the oldest to youngest in time. Each group encompasses one or more of the succeeding ones in it. As in a filing system, divisions progress from major to minor, with the smallest entry always being made under the Species (or subspecies) name, which name is found under its appropriate Genus, genus under appropriate family, and so on up to kingdom.

VERTEBRATE ANCESTRY

The following animals are often treated with fishes in the Phylum Chordata on the basis of the common characteristic of a notochord (hence termed Chordates):

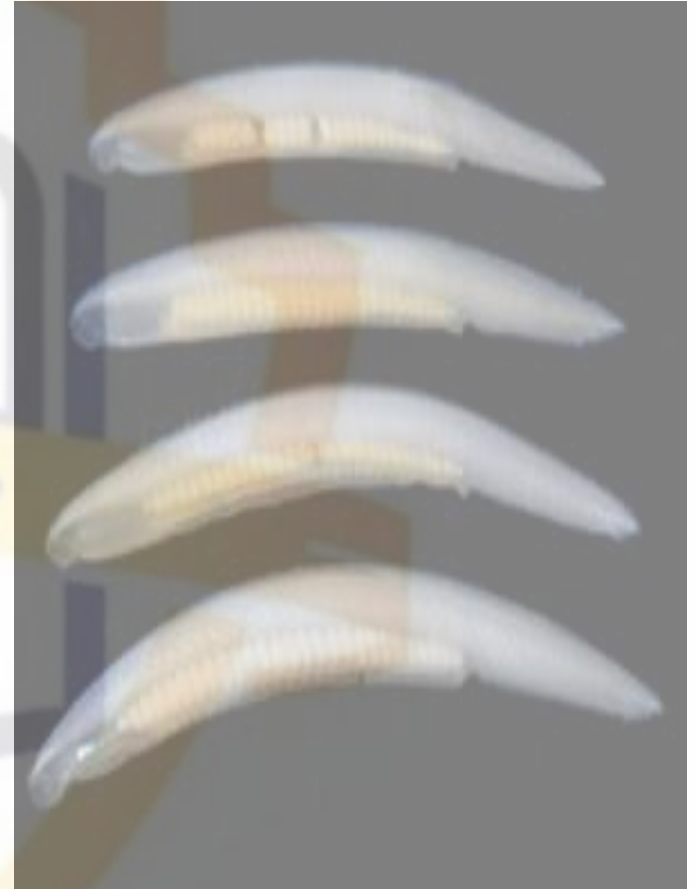
- Lacking both a cranium (Acrania) and vertebrae (Protochordata)
- Acorn worms (such as Balanoglossus)
- Sea squirts or tunicates (such as Molgula)
- Lancelets (such as amphioxus, Branchiostoma)



Acorn worms



Sea squirts or tunicates



Lancelets

Contd...

Possessing both a cranium (Craniata) and vertebrae (Vertebrata)

- Fishes (Pisces and relatives)
- Amphibians (Amphibia)-frogs, toads, salamanders, caecilians
- Reptiles (Reptilia)-lizards, snakes, turtles, crocodiles
- Birds (Aves)
- Mammals (Mammalia)

Contd...

The living chordate groups leading up to and including the fishes have Often been named as follows:

Phylum Chordata

1. Subphylum Hemichordata (acorn worms)
2. Subphylum Urochordata (sea squirts)
3. Subphylum Cephalochordata (lancelets)
4. Subphylum Vertebrata (fishes through mammals)

Contd...

- ❑ The ancestors of vertebrates were among the proto-chordates and generally without hard skeletal structures. The remote Metazoan ancestor of the protochordates and vertebrates may have been a simple sedentary creature composed of little but a digestive tract and an apparatus for gathering food particles from the surrounding water.
- ❑ The lowest chordate types, which no doubt had bottom dwelling adults and a free-swimming ciliate larva such as the tornaria of development toward a higher chordate would be the appearance certain hemichordates, possess this type of structure.
- ❑ The next stage is of gill-slit, filter-feeding, which, indeed, is found among true living vertebrates in the lamprey larva, as well as others. From the ammocoete-like Larval stage, the next step may have been to a fish-like form with direct a system development.

Table 2.1. Comparative Terminology for Major Groups that Include Living Fishes—Classes (and Subclasses)

Jordan (1923)	Regan (1929)	Berg (1940)	Traité de Zoologie (1958) ^a	Present Work
MARSIPO-BRANCHII Lampreys and hag-fishes	MARSIPO-BRANCHII	PETROMYZONTES Lampreys	CYCLOSTOMI (Cephalaspido-morphi) Lampreys (Pteraspidomorphi) Hagfishes	CEPHALASPIDO-MORPHI Lampreys and hagfishes
ELASMOBRANCHII (Selachii) Sharks, rays and skates (Holocephali) Chimaeras	SELACHII (Euselachii)	MYXINI Hagfishes ELASMOBRANCHII Sharks, rays and skates	CHONDRICHTHYES (Euselachii) Sharks, rays and skates	CHONDRICHTHYES (Elasmobranchii) Sharks, rays and skates
PISCES Bony fishes (Dipneusti) Lungfishes (Crossopterygii) Lobefins	PISCES (Crossopterygii) (Paleopterygii) Primitive bony fishes (Neopterygii) Modern bony fishes	HOLOCEPHALI Chimaeras DIPNOI TELEOSTOMI (Crossopterygii) ^b (Actinopterygii) Rayfins	(Bradyodonti) Chimaeras OSTEICHTHYES (Dipneusti) Lungfishes (Crossopterygii) Lobefins (Brachiopterygii) Reedfishes (Actinopterygii) Rayfins	(Holocephali) Chimaeras OSTEICHTHYES (Dipnoi) Lungfishes (Crossopterygii) Lobefins (Actinopterygii) Rayfins

^a From the French text of the original work, the rest.

Contd...

- ❑ Many theories of chordate evolution have been proposed and most have been discarded.
- ❑ The remarkable structural similarity between certain Ordovician echinoderms (mitrates) and heterostracans has led to a new, controversial theory of vertebrate origins.
- ❑ The mitrates are nearly symmetrical and are believed to have had a notochord, a dorsal nerve chord, and a complex central Nervous system like ostracoderms, in addition to gill slits in the pharynx like those of hemichordates.
- ❑ Early heterostracans could have evolved directly from mitrates



Ordovician echinoderms

MAJOR GROUPS OF LIVING FISHES

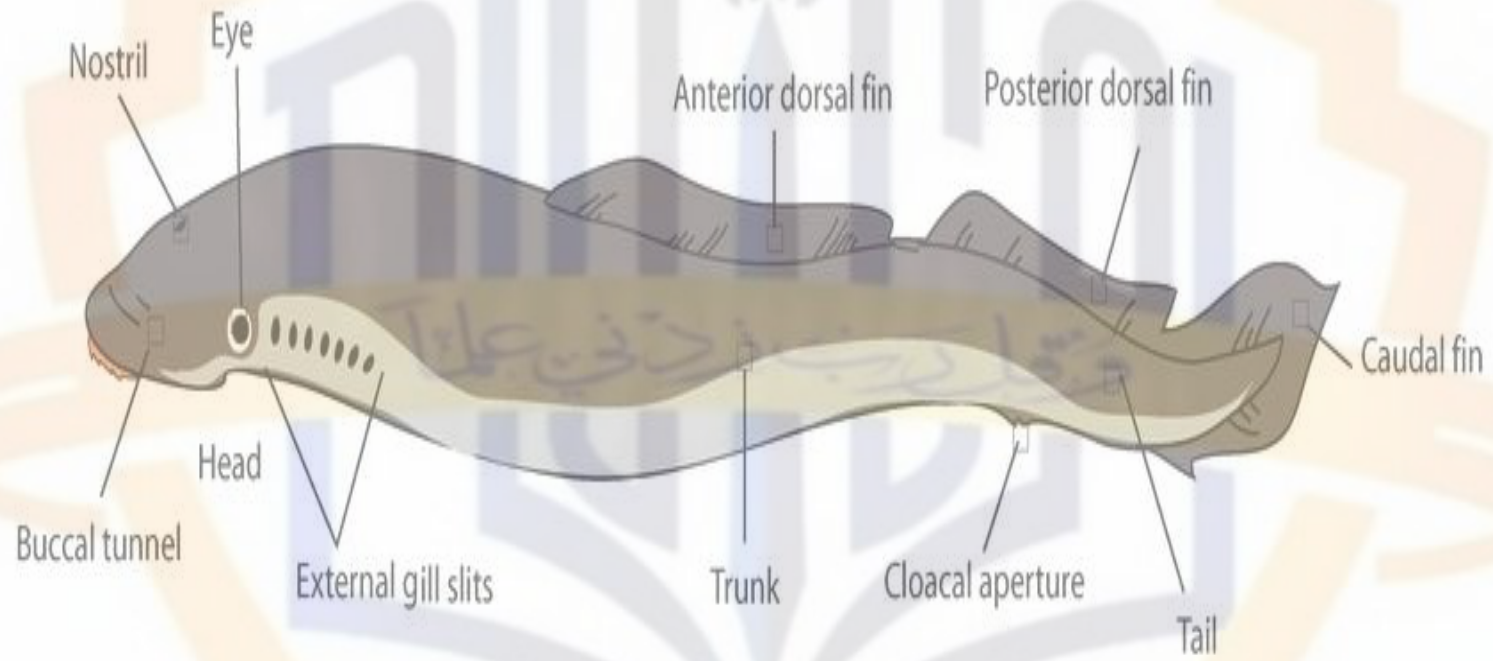
- ❑ Living fishes probably exceed 28,000 species, an abundance and diversity unequalled among all other vertebrates combined. Their ancient ancestry, extending into the past for some 500 million years, has allowed a vast span of time for evolutionary divergence and for the origin and extinction of major phyletic lines.
- ❑ Even the modern bony fishes, or teleosts, had their origin as long ago as 200 million years, in the Triassic period, when the Major groups (classes and subclasses) of fishes were firmly established.
- ❑ The Evolution and relationships of fishes are still highly debatable, despite repeated and continuing investigation; partly because of the variety of philosophical approaches used by systematists, details of the higher classification of these animals remain evanescent.

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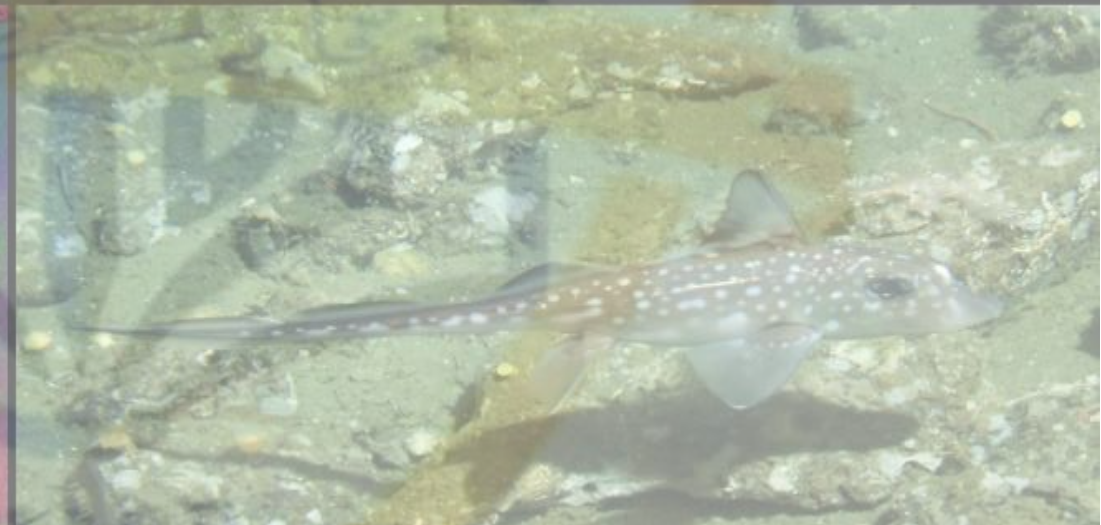
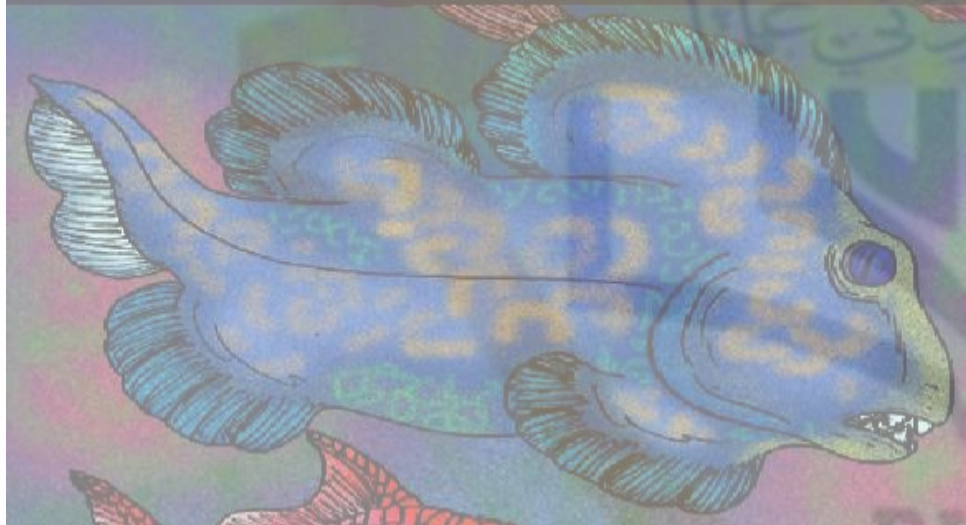
The primary Division in vertebrate classification is between the jawless vertebrates (earliest fishes), the Agnatha, and those vertebrates that possess jaws, the Gnathostomata.

We here adopt Pisces as a convenient name to include all fish and fish-like Vertebrates, and use the terms given below for the major groups of living Fishes. These constitute three important natural divisions:

- (1) **CEPHALASPIDOMORPHI**-the hagfishes and lampreys, which are jawless (agnathous) and have pouched gills;
- (2) **CHONDRICHTHYES**-the chimaeras, Sharks, rays, and their allies, which possess true jaws (gnathostomous), have Gills borne on wall-like partitions between gill chambers (a single chamber in Chimaeras), and have a cartilaginous skeleton (that may be calcified but not Bony).
- (3) **OSTEICHTHYES**-all higher fishes, which are gnathostomous, Have a branchial chamber protected by a hyoidean opercular series, including the gill cover or operculum, and have a bony skeleton.



CEPHALASPIDOMORPHI-Lamprey



CHONDRICHTHYES

Orders of Bony Fish (Class Osteichthyes)

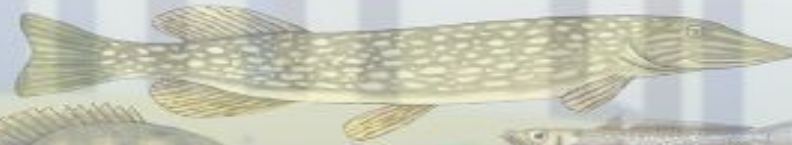


Lepisosteiformes
Gars



Anguilliformes
Eels

Perciformes (Largest Order)
Most Ocean and Freshwater Fish



Tetraodontiformes
Pufferfish and Box fish



Lophiiformes
Anglerfish



Gasterosteiformes
Seahorses and Pipefish



Elopiformes
Tarpons and Ladyfish



Siluriformes
Catfish and Bullheads



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Pisces (Groups with living representatives)

I. **Agnatha**-the jawless cyclostomes, a relict group derived from extinct ostracoderms.

1. Class Cephalaspidomorphi-comprising the Subclass Cyclostomata--Myxiniiformes, hagfishes and slime eels, and the Petromyzoniformes, Lampreys.

II. **Gnathostomata**-jawed fishes.

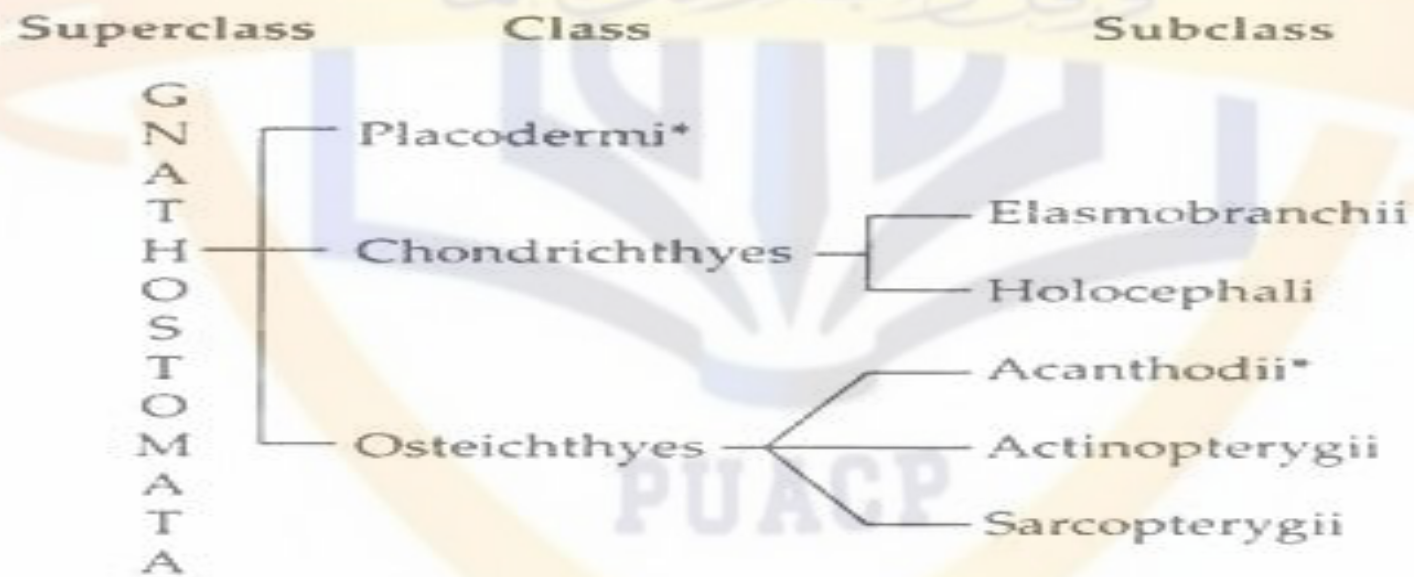
(Superclass Elasmobranchiomorphi-the extinct placoderms and the Sharks, holocephalans, and their relatives)

2. **Class Chondrichthyes**-the cartilaginous chimaeras, sharks, rays, and their allies.
(Superclass teleostomi-the fossil acanthodians and others and the bony Fishes)

3. **Class Osteichthyes**-coelacanth, lungfishes, and all other bony fishes, which include more than 90 percent of the species of living fishes.

Classification

Classificatory scheme of fishes by J. Z. Young (1981) is shown:



CHARACTERIZATION OF LIVING FISH GROUPS

- ❑ In characterizing the living representatives of the major groups of fishes, it is necessary to generalize.
- ❑ It must be understood, therefore, that in the following lists of features the most frequent condition is stated and exceptions, even where they are known, are usually omitted.
- ❑ In addition to gross characterization, useful terms that have been applied to subgroups or to Anatomical features are included.

Class Cephalaspidomorphi (Hagfishes and Lampreys)

- ☐ Notochord unconstricted.
- ☐ Jawless (agnathous).
- ☐ Main axial skeleton (vertebrae) cartilaginous or fibrous.
- ☐ Two semicircular canals in ear on each side of head in lampreys but only One on each side in hagfishes.
- ☐ No true gill arches for the support and protection of gills; instead Branchial basket situated external to the gills, gill arteries (branchial Arteries and truncus arteriosus), and branchial nerves; gills in pouches.
- ☐ Branchial basket firmly united to brain case (neurocranium).
- ☐ Paired fins (pectorals and pelvics) absent.
- ☐ Single median nostril (monorhinous).

Class Chondrichthyes (Chimaeras, Sharks, Rays, and Skates)

- ☐ Notochord constricted by vertebrae.
- ☐ Jawed (gnathostomous).
- ☐ Vertebrae cartilaginous (with some calcification but no ossification in Living forms).
- ☐ Three semicircular canals in ear on each side of head.
- ☐ Gill arches cartilaginous and internal to gills and their supplying arteries and nerves.
- ☐ Gill arches not firmly united to brain case, but joined to it by connective tissue.
- ☐ Paired fins present.
- ☐ Paired nostrils (dirhinous).

Subclass Holocephali (Chimaeras)

- ☐ Gills in four pairs; gill clefts a single pair.
- ☐ Spiracle absent.
- ☐ Skin naked in adults.
- ☐ Cloaca absent.
- ☐ Males with pelvic intromittent organs (claspers) that may have prepelvic Tenacula, and some (Chimaera) have a clasping organ (tenaculum) on forehead.



Holocephali (Chimaeras)

Subclass Elasmobranchii (Sharks, Rays, and Skates)

- ☐ Gills and gill clefts in five to seven pairs.
- ☐ Spiracle present.
- ☐ Placoid scales (dermal denticles) present or absent.
- ☐ Cloaca present.
- ☐ Males usually with pelvic intromittent organs (myxopterygia).

Class Osteichthyes (Bony Fishes)

- ☐ Notochord unconstricted or constricted.
- ☐ Jawed (gnathostomous).
- ☐ Skeleton bony.
- ☐ Three semicircular canals in ear on each side of head.
- ☐ Gill arches bony and internal to gills and their supplying arteries and Nerves.
- ☐ Gill arches not firmly united to brain case. Paired fins present.
- ☐ Paired nostrils (dirhinous).

Subclass Dipnoi (Lungfishes)

- ☐ Maxilla and premaxilla absent, three pairs of tooth plates.
- ☐ Internal nares (in recent forms).
- ☐ No movable joint between anterior and posterior part of skull.
- ☐ Palatoquadrate fused to cranium.
- ☐ Extension of radials and muscles into fin base (in recent forms); a single dorsal fin.
- ☐ Cloaca present.

Australian lungfish
(*Neoceratodus forsteri*)



African lungfish
(*Protopterus annectens*)

South American lungfish
(*Lepidosiren paradoxa*)



Devonian lungfish
(*Dipterus*)

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Lungfishes

Subclass Crossopterygii (Lobefins)

- ☐ Maxilla absent (except in some fossil forms); premaxilla present; teeth Normal.
- ☐ No internal nares (in recent forms).
- ☐ Movable joint between anterior and posterior part of skull.
- ☐ Palatoquadrate not fused to cranium.
- ☐ Extension of radials and muscles into fin base; two separate dorsal fins.
- ☐ Cloaca absent.



Devonian rhipidistian
(*Eusthenopteron foordi*)



modern coelacanth
(*Latimeria chalumnae*)

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Crossopterygii

Subclass Actinopterygii (Rayfins, Higher Bony Fishes)

- ☐ Maxilla and premaxilla present.
- ☐ No internal nares.
- ☐ No movable joint between anterior and posterior part of skull.
- ☐ Palatoquadrate not fused to cranium.
- ☐ No extension of radials and muscles into fin base (except in Polypteriformes); dorsal fin single or divided.
- ☐ Cloaca absent.



Some representatives of (A) Osteoglossiformes, (B) Clupeiformes, (C) Pleuronectiformes, (D) Ovalentaria (Polycentridae), (E) Synbranchiformes, (F) Galaxiiformes, (G) Gobiiformes, (H) Tetraodontiformes, two Perciformes (I) Perciliidae, and (J) Scianidae. Not in scale